Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1
library(tidyverse)
library(rvest)
library(glue)
library(here)

# check wd
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
my_theme <- theme_classic() +
  theme(
    legend.position = "bottom"
  )

theme_set(my_theme)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2

website <- read_html(
  "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024")
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- PWSID
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "03-32-010", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```
#3

# function to simplify reading in each element
scrape_html_text <- function(element, website){
  return(
    website %>%
      html_nodes(element) %>%
      html_text()
  )
}

# 1. System Information
water_system_name <- "div+ table tr:nth-child(1) td:nth-child(2)" %>%
  scrape_html_text(website)

pwsid <- "td tr:nth-child(1) td:nth-child(5)" %>%
  scrape_html_text(website)

ownership <- "div+ table tr:nth-child(2) td:nth-child(4)" %>%
  scrape_html_text(website)

# 3. Water Supply Sources
max_daily_use <- "th~ td+ td" %>%
  scrape_html_text(website)
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

   TIP: Use `rep()` to repeat a value when creating a dataframe.

   NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

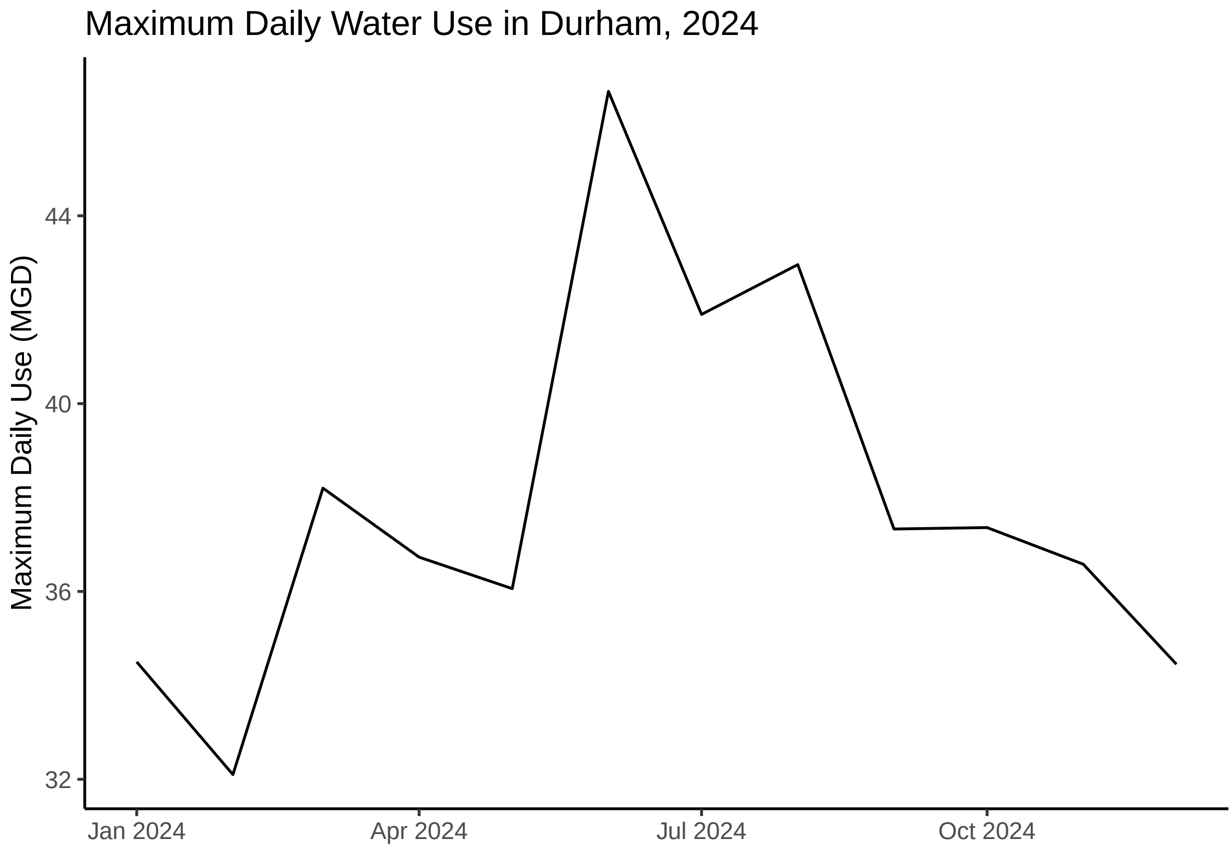
```r
#4

#months <- as.vector(rbind(1:4, 5:8, 9:12))
months <- ".fancy-table:nth-child(30) tr+ tr th" %>% scrape_html_text(website)
year <- 2024

df_water_use <- data.frame("month" = months,
                           "max_daily_use" = as.numeric(max_daily_use)) %>%
  mutate(ownership = ownership,
         pwsid = pwsid,
         water_system_name = water_system_name,
         date = ymd(paste(year, month, "1", sep = "-")))

#5

ggplot(df_water_use, aes(x = date, y = max_daily_use)) +
  geom_line() +
  labs(
    x = NULL,
    y = "Maximum Daily Use (MGD)",
    title = glue("Maximum Daily Water Use in {water_system_name}, {year}")
  )
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - "PWSID" and "year" - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.

scrape_website <- function(pwsid, year){
  url <- glue(
    "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid={pwsid}&year={year}")

  website <- read_html(url)

  # scrape relevant data
  water_system_name <- "div+ table tr:nth-child(1) td:nth-child(2)" %>%
    scrape_html_text(website)

  ownership <- "div+ table tr:nth-child(2) td:nth-child(4)" %>%
    scrape_html_text(website)

  max_daily_use <- "th~ td+ td" %>% scrape_html_text(website)
```

```
  # location of months field changes between years, using another solution
  # months <- ".fancy-table:nth-child(30) tr+ tr th" %>% scrape_html_text(website)
  months <- as.vector(rbind(1:4, 5:8, 9:12))

  df_water_use <- data.frame("month" = months,
                             "max_daily_use" = as.numeric(max_daily_use)) %>%
    mutate(ownership = ownership,
           pwsid = pwsid,
           water_system_name = water_system_name,
           date = ymd(paste(year, month, "1", sep = "-")))

  return(df_water_use)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

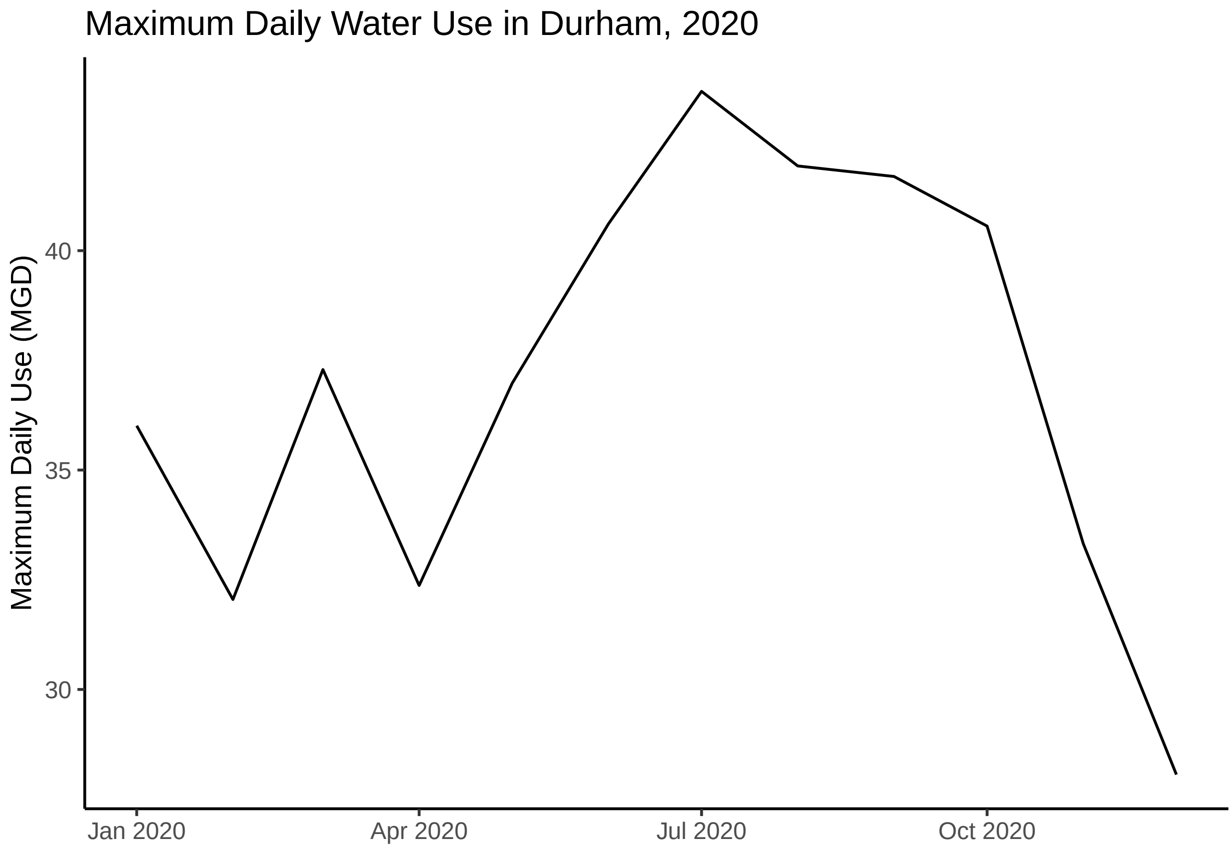
```
#7

durham_water_use_2020 <- scrape_website(pwsid = "03-32-010",
                                        year = 2020)

glimpse(durham_water_use_2020)
```

```
## Rows: 12
## Columns: 6
## $ month             <int> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ max_daily_use     <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 43.~
## $ ownership         <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ pwsid             <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", ~
## $ water_system_name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "D~
## $ date              <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02-01, 2020~
```

```
ggplot(durham_water_use_2020, aes(x = date, y = max_daily_use)) +
  geom_line() +
  labs(
    x = NULL,
    y = "Maximum Daily Use (MGD)",
    title = "Maximum Daily Water Use in Durham, 2020"
  )
```

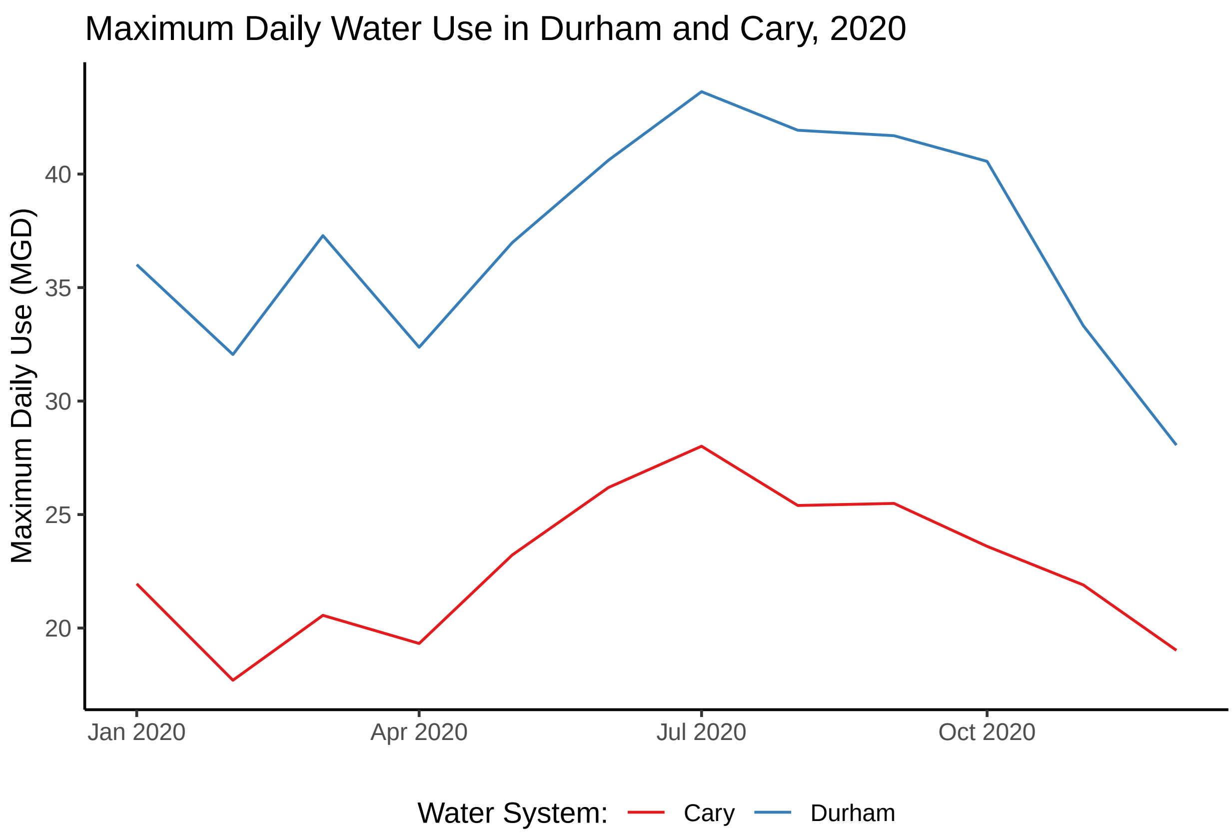


8. Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8

cary_water_use_2020 <- scrape_website(pwsid = "03-92-020", year = 2020)

bind_rows(durham_water_use_2020, cary_water_use_2020) %>%
  ggplot(aes(x = date, y = max_daily_use, color = water_system_name)) +
  geom_line() +
  scale_color_brewer(palette = "Set1") +
  labs(
    x = NULL,
    y = "Maximum Daily Use (MGD)",
    color = "Water System:",
    title = "Maximum Daily Water Use in Durham and Cary, 2020"
  )
```



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

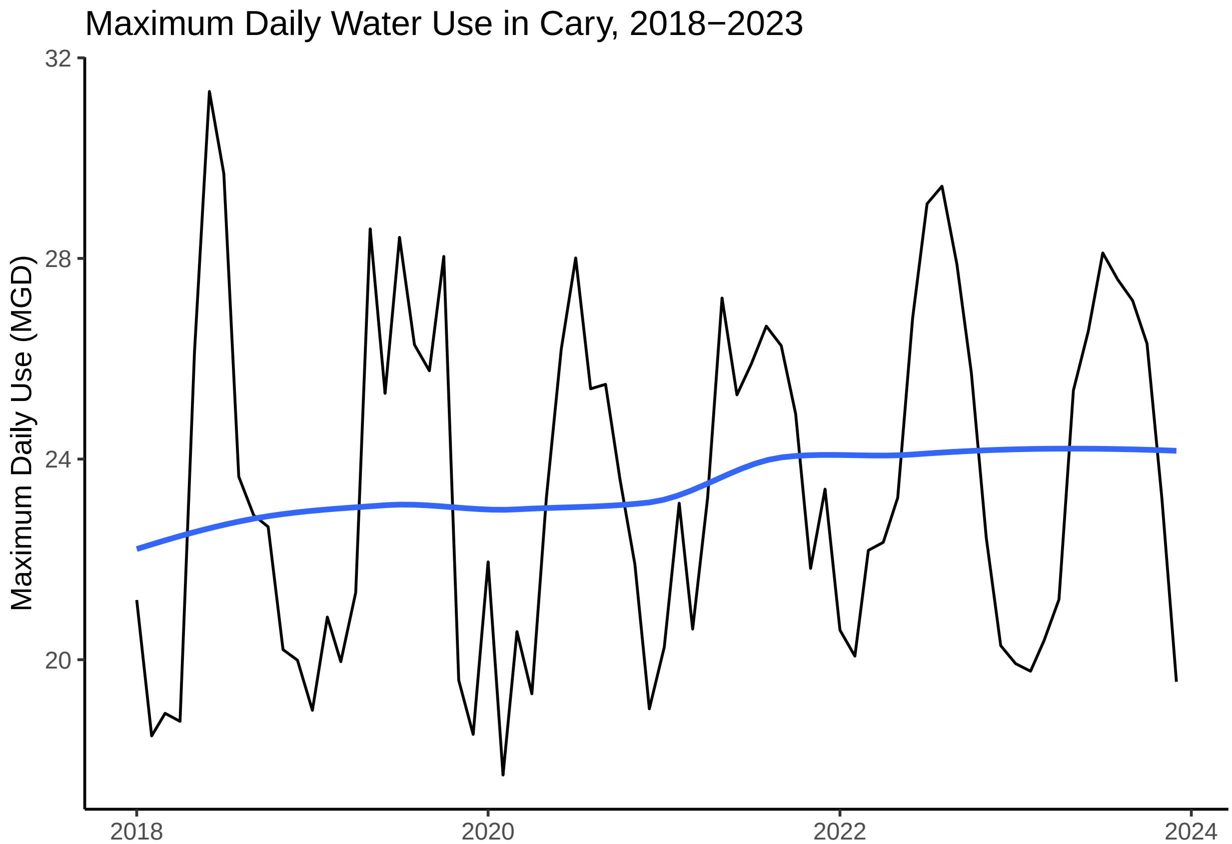
   TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```
#9

cary_water_use_2018_2023 <- c(2018:2023) %>%
  map(.f = scrape_website, pwsid = "03-92-020") %>%
  bind_rows()

ggplot(cary_water_use_2018_2023, aes(x = date, y = max_daily_use)) +
  geom_line() +
  geom_smooth(se = FALSE, method = "loess") +
  labs(
    x = NULL,
    y = "Maximum Daily Use (MGD)",
    title = "Maximum Daily Water Use in Cary, 2018-2023"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time?

Answer: The smoothed line suggests there has been a slight positive trend in water usage in Cary between 2018 and 2023. There is also a clear seasonal component to water usage.